Lab 3 – Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk System

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is organized into Presentation, Business, and Data layers. The Presentation Layer contains the User Interface Component for touch-screen interaction. The Business Layer contains the Order Manager, Payment Service, and Receipt Printer Components for ordering, payment processing, and receipt printing. The Data Layer contains the Database Component for storing menu and pricing information.

Reason 1 – Clear Separation of Concerns

Layered architecture separates customer interaction, business processing, and data storage into distinct responsibilities. The User Interface handles touch-screen interactions, while the Order Manager coordinates the ordering workflow and communicates with the payment, receipt-printing, and database components through defined interfaces.

Reason 2 – Suitable for the Kiosk Scenario

The kiosk has focused and well-defined functionality: customers select coffee and size, pay by credit card, receive a printed receipt, and use stored menu and pricing data. Separating these responsibilities makes the system easier to understand, maintain, and modify without unnecessarily affecting other parts.

Security Advantage

Payment processing is isolated within the Payment Service Component. The User Interface and Order Manager access payment functionality through the defined Payment API rather than directly accessing its implementation details. This helps restrict direct access to sensitive payment operations.

Performance Benefit

The layered structure keeps processing responsibilities organized and allows the Order Manager to communicate directly with the required services. Keeping business processing separate from touch-screen presentation reduces unnecessary work in the interface layer and helps maintain responsive customer interactions.